

AI V3 Generalized Compound Requests

Owner: AI Engineering
Status: In review
Last reviewed: 2026-08-24
PDF: [ai_v3_compound_requests.pdf](#)
Tickets: [PRO-16](#)
(parent), [PRO-18](#)
(architecture), [PRO-19](#)
(coverage)

Working record of the compound-request work: what was wrong, what changed, how Mixroom behaves now, and which live and unit tests were run. Nightcore, chipmunk, orbit, and garage are **eval prompts**, not product modes.

Architecture stays
[ADR 0002](#): one V3 planner,
Flutter prepares and executes. ChatGPT’s 13-step remix essay (import, invented drums, master, export) is not acceptance.

Problem

Vague production goals (“make this a nightcore remix”, “orbit in headphones”, “make this a garage remix”) collapsed to the nearest single command, usually `clip.align_tempo_to_project`. Named singles (“make this louder”, “pitch the vocal +3”) were already reliable. Style goals were not.

An earlier prototype injected per-genre recipes (regex + hardcoded semitones / BPM ratios). That overfit named styles and was rejected. The replacement is a **request-independent musical-dimension compiler**: the same instructions on every planner call. Flutter never matches genre names.

What changed

Area	Before	After
Style / remix / listening-format goals	Often one nearby edit (tempo-align)	One plan with every Mixroom-legal implied step: tempo, pitch, mix, pan automation; arrangement only when the user asked

Area	Before	After
Planner prompt	No always-on compiler; brief genre injection was tried and dropped	<code>aiV3MusicalDimensionCompilerInstructions</code> on every V3 request, including adaptive first-shot and continuation
Align-only collapse	First plan could ship as tempo-align only	Syntax-only retry when the request looks like a production goal (<code>remix / version / turn this into / sound like</code>) and the plan is align-only
Pan / gain automation	<code>mix:pan / mix:gain</code> often missing from context	Every audio row always exposes <code>volume , mix:gain , mix:pan</code>
Static pan vs sweep	Orbit could also emit mix-widen / static pan	If the plan writes <code>automation.set_points</code> on <code>mix:pan</code> , pan intents are stripped from <code>mix.apply_goal</code>
Pitch-only metaphors ("chipmunk", "squeaky")	Could pull a full nightcore stack	Named pitch edit only, unless the user also asked for a remix / version / sound-like goal. Defaults stay a few semitones, not an octave, unless the user names an amount
Implied vs explicit arrangement	Style goals could place library drums if assets existed	Style goals must not add parts. "Add a bassline" may create MIDI
Skip copy on named singles	Import/export/drums lecture on "make this louder"	Skip notes only on multi-command or production-shaped plans (<code>mix.apply_goal</code> , pan automation, <code>sample.place</code> , <code>row.create</code>)
Skip tense	"I'll skip" / "won't generate" after apply	Rewritten to past tense ("Mixroom skipped", "did not generate")
Chat receipt	System bullet list plus assistant sentence; mix listed every EQ/compressor knob	One assistant receipt. Mix stays an intent line. Play and Undo chips on that bubble. The floating "Applied N changes" toast is kept
Stale selection	Invalid primary clip could fail snapshot building	Invalid primary clip / clip indices are dropped instead of failing the plan

Code map

File	Role
<code>lib/ai/v3/ai_v3_style_compiler.dart</code>	Always-on compiler + align-collapse retry predicate
<code>lib/ai/v3/ai_v3_automation_targets.dart</code>	Always-on pan/gain targets; strip pan from mix when a sweep exists
<code>lib/ai/v3/ai_v3_planner_request.dart</code>	Injects compiler into the one-shot planner

File	Role
lib/ai/v3/ai_v3_adaptive_midi_planner.dart	Same compiler on adaptive first-shot and continuation
lib/ai/v3/ai_v3_planner_service.dart	One align-collapse retry
lib/ai/v3/ai_v3_preparer.dart	Mix pan-intent strip at prepare time
lib/ai/v3/ai_v3_context.dart , ai_v3_planning_snapshot.dart	Automation targets + stale selection
lib/ai/chat_pipeline.dart	Receipt text, skip attach, skip tense, mix-detail collapse
lib/screens/audio_editor.dart	One assistant bubble; Play (seek 0 + play) and Undo chips

Behavior: before vs after

Project under test: one vocal/instrumental audio clip unless noted.

Live runs used macOS Mixroom with

```
flutter run -d macos --dart-define-from-file=tool/local_ai_debug.json .
```

Prompt	Before	After (live)
make this louder	Gain up; sometimes skip lecture	+3 dB only. One receipt. No skip lecture
chipmunk this	Could become full nightcore (tempo + mix)	Pitch only (live: +12). Compiler now asks for a moderate default
orbit in headphones	Missing pan target, or mix-widen instead of a sweep	10 pan points , no mix-widen
make this a nightcore remix	Tempo-align only, or stacked extra pitch	150 BPM, +3 pitch, brighter EQ , one row. No import/drums/export
make this a garage remix	Invented library drums; future-tense skip copy	132 BPM + compressor/EQ , no drums. Skip copy past tense
add a bassline	Inconsistent	MIDI row + 32-note clip (explicit arrangement)
add nightcore drums	Placed library loops	Still out of product: Mixroom does not invent drum parts. Must say so
what is nightcore?	Should not mutate	Unchanged: question, no edits

Unsupported on purpose (say so, do not fake it): import, export, generated drums, true binaural 8D / HRIR.

Experiments

Lab 1 — Pre-fix live (2026-08-21, ~17:54–18:07)

16 prompts against a live project. **13 pass, 2 partial, 1 fail (88%).**

- Fail: explicit “add nightcore drums” placed library loops.
- Partial: chipmunk → full nightcore stack.
- Partial: slowed + reverb stacked extra pitch.

Lab 2 — Post-compiler live (2026-08-21, ~19:03–19:11)

Six held-out / regression prompts. **6/6 pass.**

Louder, chipmunk, orbit, nightcore, garage, add bassline — results in the table above.

Lab 3 — UX pass (code complete; Play/Undo live check still pending)

After Lab 2, chat still duplicated receipts and dumped mix knobs. That pass added:

1. One assistant receipt (system execution list removed on V3 success).
2. Mix `mix.apply_goal` details collapsed to the receipt intent.
3. Broader past-tense skip rewriter (`won't generate`, `I'll skip`, `will skip`).
4. Moderate pitch-only metaphor wording in the compiler.
5. Play and Undo chips on the verified completion bubble.

The bottom toast “Applied N changes” was left in place by product choice.

Play/Undo chips are in `audio_editor.dart` but have not been clicked in a fresh live session since they landed. Hot-reload Mixroom and tap them once on a verified apply.

Automated tests

Use `flutter test`, not `dart test`.

```
flutter test \
  test/ai_v3_style_compiler_test.dart \
  test/ai_v3_automation_targets_test.dart \
  test/ai_v3_planning_snapshot_test.dart \
  test/ai_v3_context_test.dart \
  test/ai_v3_contract_test.dart \
  test/ai_v3_planner_service_test.dart \
  test/assistant_action_flow_test.dart
```

Coverage includes: compiler has no genre names or hardcoded +3 / 1.25x; nightcore / phonk / set-tempo / questions share identical instructions; align-collapse retries only goal-shaped align-only plans; pan targets always present; mix pan stripped when a sweep exists; skip notes omitted on named singles; skip copy past-tensed; mix execution details collapse to intent.

Manual integration checklist

Replay on macOS with a project that already has audio:

1. `make this louder` — gain only; one chat bubble; toast OK.
2. `chipmunk this` — pitch only.
3. `orbit in headphones` — pan automation, no widen.
4. `make this a nightcore remix` — faster + higher + brighter; no new drums.
5. `make this a garage remix` — groove/mix; no invented drums; past-tense skip.
6. `add a bassline` — may create MIDI.
7. `what is nightcore?` — no mutation.
8. Tap **Play** on the receipt — playhead to start, playback.
9. Tap **Undo** — reverts the last AI change while that entry is still on top.

Out of scope

- ChatGPT import / invented drums / true 8D / master / export.
- Genre regex catalogs and hardcoded BPM/pitch recipes.
- [PRO-51](#) project-open hang.
- Linear status moves (tickets remain In Progress until review).

Update trigger

Update this page when the compiler instructions, align-collapse retry, automation-target guarantee, mix pan-strip, skip-note policy, or verified chat receipt / Play / Undo behavior changes.